DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Data Structures and Applications | Course Code: 21CS32 | Credits: 4

Exam Session: CIA-1 Midterm 2024 **Max Marks: 80 | Duration: 3 Hours**

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Explain AVL Tree rotations (LL, RR, LR, RL) with balance factor criteria and demonstrate insertion of keys: [14, 17, 11, 7, 53, 4, 13] into an empty AVL tree. **[8 Marks]**

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Q2. Write an algorithm and C function to convert an Infix expression to Postfix using a Stack. Trace the algorithm on: (A + B * C) / (D - E ^ F). **[8 Marks]**

MODULE 2

Q3. Explain Dijkstra’s Single Source Shortest Path algorithm on a directed weighted graph and write the pseudocode with time complexity using adjacency list and min-priority queue. **[8 Marks]**